

CLEAN ENERGY IMPACT QUANTIFICATION & MRV

Impact Quantification Report

CNG Fleet Conversion

Twelve-month verified impact assessment of a 40-vehicle compressed natural gas conversion program – fuel displacement, avoided emissions, and financial performance.

CLIENT

Meridian Logistics
Nigeria Ltd

REPORTING PERIOD

July 2025 -
June 2026

SCOPE

40 vehicles
Lagos-Ibadan-Abuja

REPORT DATE

August 2026
v1.0 - Final

This is a sample report

CircularEnergies prepared this document to demonstrate our Clean Energy Impact Quantification & MRV methodology as applied to vehicle fleet conversion. **Meridian Logistics Nigeria Ltd is a fictional entity.** All operating data, fuel figures, emission calculations and financial results are simulated to reflect realistic performance for a mixed 40-vehicle commercial fleet converted to CNG in southwestern Nigeria.

Live engagements use actual telematics and fuel-card data, site-specific fuel properties, and produce reports suitable for ESG investor disclosure, DFI results framework submission, and carbon credit pre-feasibility assessment.

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Why fleet conversion is measured differently

Impact quantification for a vehicle fleet differs from a stationary energy installation in four material respects, each of which shapes this report's structure:

- **Emissions are Scope 1, not Scope 2.** Fuel combusted in owned or controlled vehicles is a direct emission. Facility solar displaces purchased electricity and reports under Scope 2. The disclosure language in Section 08 reflects this.
- **The counterfactual is fuel-specific.** A bi-fuel petrol conversion and a dual-fuel diesel retrofit displace different fuels at different substitution rates. They must be modelled as separate segments and only aggregated at the reporting layer.
- **Methane slip must be deducted.** Unburned methane passing through the engine is a potent greenhouse gas. Reports that omit this adjustment systematically overstate impact – and will not survive DFI or verifier scrutiny.
- **The primary value is economic, not carbon.** As Section 07 shows, CNG conversion delivers a large fuel cost reduction and a comparatively modest emissions reduction. We report both honestly rather than emphasizing one.

01 Executive summary

Over the twelve months to June 2026, Meridian Logistics operated a converted 40-vehicle fleet across 2,460,960 kilometres of commercial service. The programme displaced 142,448 litres of petrol and 169,920 litres of diesel, substituting 331,872 standard cubic metres of compressed natural gas.

DISTANCE OPERATED

2.46M

vehicle-kilometres, 40 units

LIQUID FUEL DISPLACED

312k L

petrol and diesel combined

NET EMISSIONS AVOIDED

78.5 t

tCO₂e, after methane adjustment

NET FUEL COST SAVING

₦269.0M

71% below baseline spend

Headline finding - two results, reported separately

Financial: Net fuel cost saving of ₦269,000,913 against a baseline liquid-fuel spend of ₦378,962,598 – a **71% reduction** after CNG purchase, incremental maintenance and cylinder recertification reserves. Programme capital expenditure of ₦74,200,000 was recovered in **3.3 months**.

Environmental: Gross tank-to-wheel avoided emissions of 163.84 tCO₂e (20.9% of baseline). After deducting 85.30 tCO₂e for methane slip, **net avoided emissions are 78.54 tCO₂e - a 10.0% reduction**. This is the figure we recommend for external disclosure.

A note on proportionality. The financial and environmental results are deliberately not presented as a single blended headline. CNG conversion is a strong cost-reduction measure and a moderate decarbonization measure. Fleet operators, lenders and ESG investors are best served by seeing that asymmetry stated plainly. Any advisor presenting a 71% cost saving alongside an implied equivalent carbon saving is misrepresenting the technology.

Carbon intensity

Baseline – liquid fuel only

0.496 kgCO₂e / km

Post-conversion – net of methane slip

0.464 kgCO₂e / km

Fleet-wide carbon intensity fell by 6.4% across all 40 vehicles, including the residual liquid-fuel share that CNG does not displace.

02 Fleet profile & conversion scope

The converted fleet comprises two technically distinct segments. Each was modelled separately throughout the calculation chain, because conversion type, substitution rate and displaced fuel all differ.

PARAMETER	SEGMENT A - BI-FUEL	SEGMENT B - DUAL- FUEL
Vehicle class	Light commercial vans and minibuses	Rigid and articulated trucks
Units converted	28	12
Baseline fuel	Petrol (PMS)	Diesel (AGO)
Conversion technology	Sequential bi-fuel CNG kit	Diesel dual-fuel retrofit
Mean daily distance	165 km · 26 days/month	295 km · 24 days/month
Baseline fuel economy	8.5 km/litre	3.3 km/litre
Achieved substitution rate	84%	55%
Annual distance	1,441,440 km	1,019,520 km
Liquid fuel displaced	142,448 L petrol	169,920 L diesel
Conversion capex	₹23,800,000	₹50,400,000

On substitution rates. Segment A achieves 84% CNG substitution because a sequential bi-fuel system runs almost entirely on gas once the engine reaches operating temperature, reverting to petrol only for cold start and when cylinders are depleted. Segment B is capped at 55% because a dual-fuel diesel engine requires a continuous diesel pilot injection to achieve compression ignition – the diesel cannot be fully eliminated. These are measured outcomes from fuel-card and telematics reconciliation, not manufacturer claims.

Boundary and exclusions

- **Included:** tank-to-wheel combustion emissions from all fuels consumed in the 40 converted vehicles; methane slip from CNG operation; incremental maintenance attributable to conversion.
- **Excluded:** upstream well-to-tank emissions for all fuels; embodied emissions in conversion kits and cylinders; unconverted fleet vehicles; emissions from the CNG refueling station itself, which is third-party owned and operated.
- **Rationale:** a tank-to-wheel boundary is the standard basis for GHG Protocol Scope 1 mobile combustion reporting. Section 09 notes where a well-to-wheel boundary would alter conclusions.

03 Methodology & data sources

COMPONENT	SOURCE / STANDARD	VALUE APPLIED
Petrol emission factor	IPCC 2006 Guidelines, Vol. 2 Ch. 3, Tier 1	2.31 kgCO ₂ /L
Diesel emission factor	IPCC 2006 Guidelines, Vol. 2 Ch. 3, Tier 1	2.68 kgCO ₂ /L
CNG emission factor	IPCC 2006 Guidelines, natural gas	2.75 kgCO ₂ /kg
CNG density	Nigerian pipeline gas specification	0.68 kg/SCM
Petrol energy equivalence	Lower heating value ratio	1 kg CNG $\equiv$ 1.50 L petrol
Diesel energy equivalence	Lower heating value ratio	1 kg CNG $\equiv$ 1.30 L diesel
Methane fraction in CNG	Gas supplier composition certificate	90% by mass
Methane slip rate	Conservative retrofit assumption	1.5% of throughput
Methane GWP (100-yr)	IPCC AR5	28 $\times$ CO ₂
Reporting protocol	GHG Protocol Corporate Standard	Scope 1 - mobile combustion

Data collection and quality control

- **Distance:** vehicle telematics, monthly aggregation, cross-checked against dispatch records. Variance tolerance $\pm 2\%$.
- **CNG purchased:** station dispensing receipts reconciled to fleet card statements, vehicle by vehicle.
- **Liquid fuel purchased:** fuel card transaction data, matched to vehicle registration.
- **Displacement calculation:** derived from measured CNG consumption using energy equivalence, then validated against the observed decline in liquid fuel purchases versus the pre-conversion twelve-month baseline.

Data quality flag M1 is applied throughout Section 04 and denotes: primary metered or transactional data, reconciled across two independent sources, with no estimation or gap-filling. Where a live engagement encounters missing data, lower-tier flags (M2 estimated, M3 extrapolated) are applied at line level and disclosed.

04 Monthly operating data

Twelve months of reconciled operating data. Seasonal variation reflects reduced haulage activity during the peak rainy season and increased volumes in the final quarter of the calendar year.

MONTH	DISTANCE (KM)	CNG (SCM)	PETROL DISPL. (L)	DIESEL DISPL. (L)	TCO ₂ E AVOIDED	NET SAVING (₹)	FLAG
Jul 2025	201,799	27,214	11,681	13,933	6.44	22,058,075	✓ M1
Aug 2025	194,416	26,218	11,253	13,424	6.20	21,251,072	✓ M1
Sep 2025	211,643	28,541	12,251	14,613	6.75	23,134,079	✓ M1
Oct 2025	206,721	27,877	11,966	14,273	6.60	22,596,077	✓ M1
Nov 2025	214,104	28,873	12,393	14,783	6.83	23,403,079	✓ M1
Dec 2025	194,416	26,218	11,253	13,424	6.20	21,251,072	✓ M1
Jan 2026	184,572	24,890	10,684	12,744	5.89	20,175,068	✓ M1
Feb 2026	189,494	25,554	10,969	13,084	6.05	20,713,070	✓ M1
Mar 2026	206,721	27,877	11,966	14,273	6.60	22,596,077	✓ M1
Apr 2026	216,564	29,205	12,535	14,953	6.91	23,672,080	✓ M1
May 2026	226,408	30,532	13,105	15,633	7.23	24,748,084	✓ M1
Jun 2026	214,104	28,873	12,393	14,783	6.83	23,403,079	✓ M1
TOTAL - 12 months	2,460,960	331,872	142,448	169,920	78.54	269,000,913	-

Monthly avoided emissions are stated net of methane slip. Net saving is stated after CNG purchase cost, incremental maintenance accrual and cylinder recertification reserve, allocated on a distance-weighted basis.

1

Segment A - baseline petrol consumption

$$1,441,440 \text{ km} \div 8.5 \text{ km/L}$$

= 169,581 litres petrol

2

Segment A - petrol displaced at 84% substitution

$$169,581 \text{ L} \times 0.84$$

= 142,448 litres displaced

3

Segment B - baseline diesel consumption

$$1,019,520 \text{ km} \div 3.3 \text{ km/L}$$

= 308,945 litres diesel

4

Segment B - diesel displaced at 55% substitution

$$308,945 \text{ L} \times 0.55$$

= 169,920 litres displaced

5

CNG required - energy equivalence basis

$$(142,448 \div 1.50) + (169,920 \div 1.30)$$

= 94,965 + 130,708 = 225,673 kg CNG

6

Conversion to volumetric units

$$225,673 \text{ kg} \div 0.68 \text{ kg/SCM}$$

= 331,872 SCM dispensed

Validation. Calculated CNG requirement of 331,872 SCM was reconciled against station dispensing records for the period. Agreement within tolerance confirms that the assumed substitution rates and energy equivalence ratios reflect actual fleet behaviour rather than theoretical values.

06 Greenhouse gas calculation chain

Every figure below is traceable to a metered input and a published emission factor. This is the section a verifier, lender or ESG auditor will interrogate first.

STEP	CALCULATION	RESULT
Baseline - petrol	142,448 L × 2.31 kgCO ₂ /L	329.06 tCO ₂
Baseline - diesel	169,920 L × 2.68 kgCO ₂ /L	455.39 tCO ₂
Total baseline emissions	Displaced fuel, counterfactual case	784.44 tCO₂
Project - CNG combustion	225,673 kg × 2.75 kgCO ₂ /kg	620.60 tCO ₂
Gross avoided (tank-to-wheel)	784.44 – 620.60	163.84 tCO₂
Methane throughput	225,673 kg × 90% CH ₄	203,106 kg
Methane slipped	203,106 kg × 1.5%	3,047 kg CH ₄
Slip in CO ₂ equivalent	3,047 kg × GWP 28	85.30 tCO _{2e}
NET AVOIDED EMISSIONS	163.84 – 85.30	78.54 tCO_{2e}

Why the methane adjustment matters. Methane slip reduces the reported benefit of this programme by 52% - from 163.84 to 78.54 tCO_{2e}. An impact report that ignores slip would overstate avoided emissions by a factor of 2.1×. Because slip is a function of engine technology and calibration rather than fuel volume, it is also the single most improvable variable in the programme: a reduction in slip from 1.5% to 0.75% would raise net avoided emissions to approximately 121 tCO_{2e} without any change in fleet operations.

07 Financial impact analysis

LINE	BASIS	AMOUNT (₹)
Petrol cost avoided	142,448 L × ₹1,050/L	149,570,598
Diesel cost avoided	169,920 L × ₹1,350/L	229,392,000
Gross fuel cost avoided	Baseline liquid fuel spend	378,962,598
Less: CNG purchased	331,872 SCM × ₹300/SCM	(99,561,684)
Less: incremental maintenance	₹180,000 × 40 vehicles	(7,200,000)
Less: cylinder recertification reserve	Accrual toward 3-year test cycle	(3,200,000)
NET FUEL COST SAVING	71.0% of baseline spend	269,000,913

CONVERSION CAPEX

₹74.2M

40 vehicles, both segments

SIMPLE PAYBACK

3.3 mo

on net saving basis

SAVING PER VEHICLE

₹6.7M

annual, fleet average

SAVING PER KM

₹109

across all converted units

Payback is a function of utilization, not technology. The 3.3-month payback reported here reflects a fleet running 61,524 km per vehicle per year — high-intensity commercial haulage. The same conversion on a fleet averaging 25,000 km annually would return a payback of roughly 10 to 14 months. Operators should not transfer this figure to their own fleet without recalculating against actual distance travelled. This is the single most common error we see in vendor-supplied CNG business cases.

Carbon market value

At an indicative voluntary carbon market price of US\$8.50/tCO₂e and an exchange rate of ₹1,580/US\$, the net avoided emissions of 78.54 tCO₂e carry a gross indicative value of **₹1,054,729** – equivalent to 0.39% of the fuel cost saving.

This figure is presented for completeness, not as a recommendation. At this scale, carbon revenue would not cover the cost of validation, verification and registry fees. Section 10 addresses the conditions under which a credit pathway becomes economically rational.

The statements below are drafted for direct inclusion in an annual report, sustainability report or investor disclosure pack. They are written to GHG Protocol conventions and require no further editing beyond entity naming.

GHG Protocol - Scope 1 mobile combustion

During the reporting period 1 July 2025 to 30 June 2026, the Company converted 40 commercial vehicles to compressed natural gas operation. The conversion displaced 142,448 litres of petrol and 169,920 litres of diesel, substituting 331,872 standard cubic metres of compressed natural gas. Net Scope 1 emissions avoided, calculated on a tank-to-wheel basis using IPCC 2006 Tier 1 emission factors and inclusive of a 1.5% methane slip adjustment applied at AR5 100-year global warming potential, were 78.54 tonnes of carbon dioxide equivalent. Fleet carbon intensity declined from 0.496 to 0.464 kgCO₂e per vehicle-kilometre.

TCFD / IFRS S2 metric alignment

DISCLOSURE REQUIREMENT	REPORTED VALUE
Scope 1 emissions avoided (absolute)	78.54 tCO ₂ e
Emissions intensity – physical basis	0.464 kgCO ₂ e/vehicle-km
Intensity reduction vs baseline	6.4%
Transition risk – fuel price exposure	71% reduction in fuel cost volatility exposure
Capital deployed toward transition	₦74,200,000
Methodology and assurance basis	IPCC Tier 1; primary metered data; third-party prepared

Applicable frameworks

GHG Protocol Corporate Standard

IPCC 2006 Guidelines

IFRS S2 / TCFD

GRI 302 Energy

GRI 305 Emissions

CBN Sustainable Banking Principles

NGX Sustainability Disclosure Guidelines

09 DFI results framework outputs

Indicators below are populated in the format required by the principal development finance institutions active in Nigerian transport and logistics lending.

INSTITUTION	INDICATOR	REPORTED VALUE
African Development Bank	Greenhouse gas emissions reduced (tCO ₂ e/yr)	78.54
African Development Bank	Private sector capital mobilised (US\$)	46,962
IFC - CAFI	Annual GHG emissions avoided (tCO ₂ e)	78.54
IFC - CAFI	Climate finance eligible investment (US\$)	46,962
FCDO	Number of assets converted to lower-carbon fuel	40
FCDO	Annual fuel cost saving to beneficiary (US\$)	170,254
Nigeria - PCNGi	Vehicles converted under programme	40
Nigeria - PCNGi	Litres of imported liquid fuel displaced	312,368
Nigeria NDC	Transport sector mitigation contribution (tCO ₂ e)	78.54

Boundary caveat for DFI submission. Values above are tank-to-wheel. Several DFI results frameworks are moving toward well-to-wheel accounting for fuel-switching projects, which would incorporate upstream methane emissions from gas production and compression. Under a well-to-wheel boundary the net benefit of CNG conversion narrows materially and, at high upstream leakage rates, can approach parity with diesel. We recommend disclosing the boundary explicitly in any submission rather than reporting the headline figure alone.

Applicable methodology

The relevant approved methodology for this project type is **CDM AMS-III.S** – *Introduction of low-emission vehicles and technologies to commercial vehicle fleets* – which remains available under Gold Standard and is transitionable to Article 6.4 arrangements. Verra's VCS programme does not currently offer a directly applicable methodology for CNG retrofit of existing commercial fleets.

Eligibility assessment

CRITERION	STATUS	NOTE
Project type eligible under AMS-III.S	Met	Commercial fleet, fuel-switch retrofit
Metered, auditable activity data	Met	Telematics and dispensing records, flag M1
Methane slip accounted	Met	Required under methodology; applied at 1.5%
Additionality demonstration	Not met	3.3-month payback indicates the project is financially attractive without carbon revenue
Scale economics	Not met	₦1,054,729 gross annual value against validation and verification costs typically exceeding ₦25M

Recommendation: do not pursue registration at current scale. Two independent barriers apply. First, the programme fails the additionality test — a 3.3-month payback means the conversion is plainly viable on its own economics, and a validator would reject the argument that carbon revenue was decisive. Second, even if additionality were established, gross credit value of ₦1,054,729 annually would not recover validation, verification and registry costs. A pathway becomes worth revisiting at approximately 400 to 500 converted vehicles, or where conversion is bundled into a programmatic activity aggregating multiple operators under a single crediting programme.

CircularEnergies takes the position that advising a client to pursue an uneconomic or unregistrable carbon pathway to generate advisory fees is a conflict of interest. Where the answer is no, we say so.

Assurance status

This report has been prepared by CircularEnergies as an independent third party using data supplied by the client. It has **not** been subjected to external limited or reasonable assurance under ISAE 3410. Where a lender or investor requires assured figures, this report is structured to support an assurance engagement without restructuring.

Principal limitations

- **Methane slip is assumed, not measured.** The 1.5% rate is a conservative literature value for retrofit systems. Actual slip varies with engine calibration, load profile and maintenance state, and can range from below 0.5% to above 3%. Direct exhaust measurement on a sample of vehicles would materially tighten this estimate and is the highest-value refinement available to this programme.
- **Tank-to-wheel boundary excludes upstream emissions.** Gas production, processing and compression emissions are outside the reporting boundary. This is standard for Scope 1 reporting but material to any well-to-wheel comparison.
- **Substitution rates are period averages.** Individual vehicle performance varies with route profile and driver behavior. Per-vehicle reporting is available where required.
- **Fuel prices are period averages.** Nigerian petrol, diesel and CNG prices moved materially during the reporting period. Financial results are sensitive to the CNG-to-liquid-fuel price spread, which is a policy-dependent variable.
- **Embodied emissions excluded.** Manufacture of conversion kits, cylinders and associated components is not accounted. At programme scale this is immaterial relative to operational emissions but would be included in a full lifecycle assessment.

Sensitivity - net avoided emissions

METHANE SLIP ASSUMPTION	NET AVOIDED (TCO ₂ e)	VARIANCE VS BASE CASE
0.50% - best-in-class calibration	135.40	+72%
1.00%	106.97	+36%
1.50% - base case applied	78.54	-
2.50% - poorly maintained fleet	21.67	-72%
3.00% - upper literature bound	-6.77	-109%

At a 3.0% slip rate the programme's net climate benefit falls to -6.77 tCO₂e - approximately -0.9% of baseline. This is the reason we recommend direct slip measurement rather than continued reliance on a literature assumption. The financial case is unaffected by slip; the environmental case is highly sensitive to it.

1

Commission direct methane slip measurement

Exhaust sampling across a representative sample of 6 to 8 vehicles spanning both segments. This is the highest-value action available: it converts the single largest uncertainty in this report into a measured value, and if actual slip is below the assumed 1.5%, it increases defensible reported impact at no operational cost.

2

Prioritize Segment B substitution rate improvement

Segment B operates at 55% substitution against a technical ceiling of roughly 70% for dual-fuel systems. Recalibration and driver training targeting a 5-point improvement would displace approximately 15,447 additional litres of diesel annually, worth around ₦20,853,818.

3

Do not pursue carbon credit registration at current scale

Both additionality and economics fail. Revisit only at 400+ vehicles or under a programmatic aggregation across multiple operators.

4

Disclose the boundary explicitly in all external reporting

Tank-to-wheel figures are correct for Scope 1 but will be challenged by sophisticated ESG analysts. Stating the boundary pre-empts the challenge and protects credibility.

5

Extend conversion to remaining fleet on a payback-ranked schedule

Rank unconverted vehicles by annual distance and convert highest-mileage units first. Payback is driven by kilometres travelled, not vehicle count – converting a low-utilization vehicle ties up capital at a fraction of the return.

6

Monitor the CNG-to-liquid-fuel price spread as a policy risk

The entire financial case rests on the price differential between CNG and liquid fuels, which is shaped by domestic gas pricing policy. A material narrowing of the spread would extend payback on future conversions. This belongs on the risk register, not in the assumptions.

CircularEnergies is an energy transition feasibility and intelligence advisory firm headquartered in Abuja, Nigeria. We produce bankable analysis for energy projects across Nigeria and West Africa — work that is designed to withstand scrutiny from lenders, development finance institutions and ESG investors.

Impact quantification services

- **One-time Impact Quantification Report** - full MRV assessment of a completed installation or conversion programme
- **Annual Impact Monitoring Report** - recurring quantification for continuous disclosure
- **DFI Results Framework Package** - indicators populated to institution-specific templates
- **Carbon Credit Pre-Registration Assessment** - eligibility, methodology selection and pathway costing

Adjacent capability

- **Bankable feasibility studies** - CNG, fleet conversion, solar-plus-storage, biogas
- **Project finance modelling** - sculpted debt sizing, DSCR and LLCR analysis, sensitivity suites
- **Energy risk screening** - for banks, DFIs and insurers
- **Transition readiness diagnostics** - facility and fleet

Commission an Impact Quantification Report

Every engagement begins with a scoping call at no cost. If we do not believe an impact quantification exercise would add material value to your business, we will say so directly rather than sell you a report.

info@circularenergies.africa · +234 815 824 2934 · [circularenergies.africa](https://www.circularenergies.africa)